This work sample is a group project completed as part of a Master's level course "GIS Applications in Social Science" at UChicago Harris. It analyses the association of potholes distribution in Chicago with socioeconomic factors and visualize those factors across neighborhoods.

Original GIS code files are unavailable due to loss of access to the remote server where they were stored.

Yuki's contribution included research design, data cleaning of raw administrative data using R, data analysis and visualization using ArcGIS (Slide 13-14, 21-22).

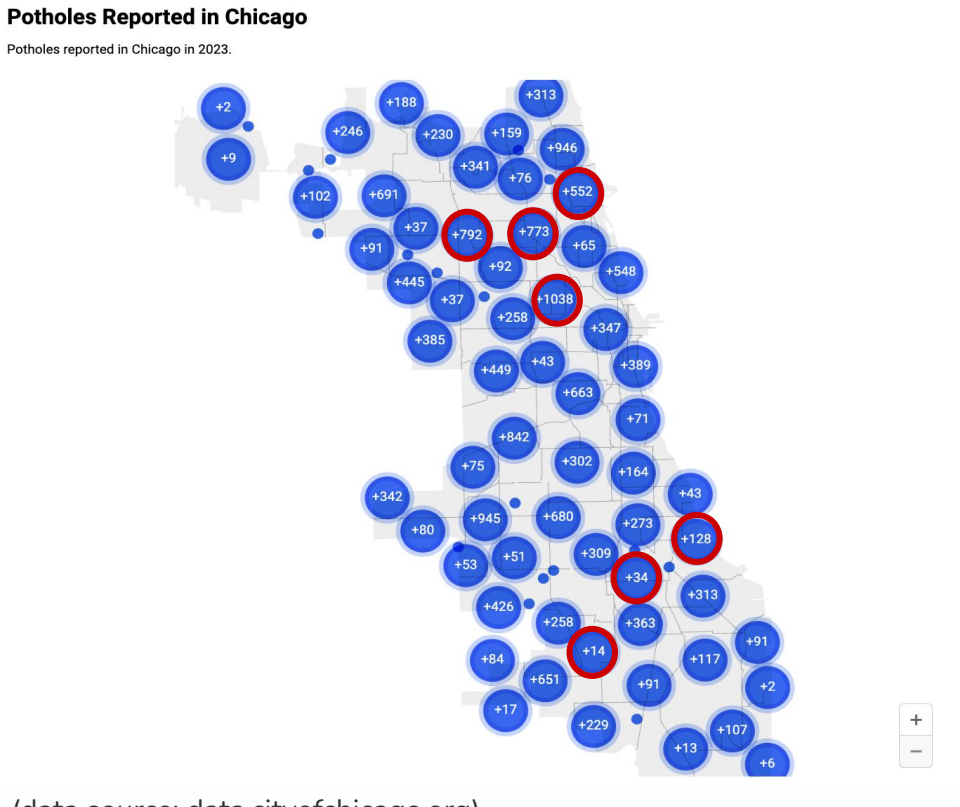
A Spatial Analysis of Potholes in Chicago

Presenters: Yuki Chen, Bruno Xie, Serena Huang

Presentation Outline

- Background
- Overview of data and research methods
- Analysis of potholes reported and patched
- Analysis of socioeconomic status of Chicago
- Analysis of spatial correlation
- Limitation & Next Steps

Potholes Reported in Chicago 2023



(data source: data.cityofchicago.org)

Potholes reported in Chicago show disparities in different area.

WHY?

Example: Road Pavement Maintenance

The Washington Post
Democracy Dies in Darkness

Sign in

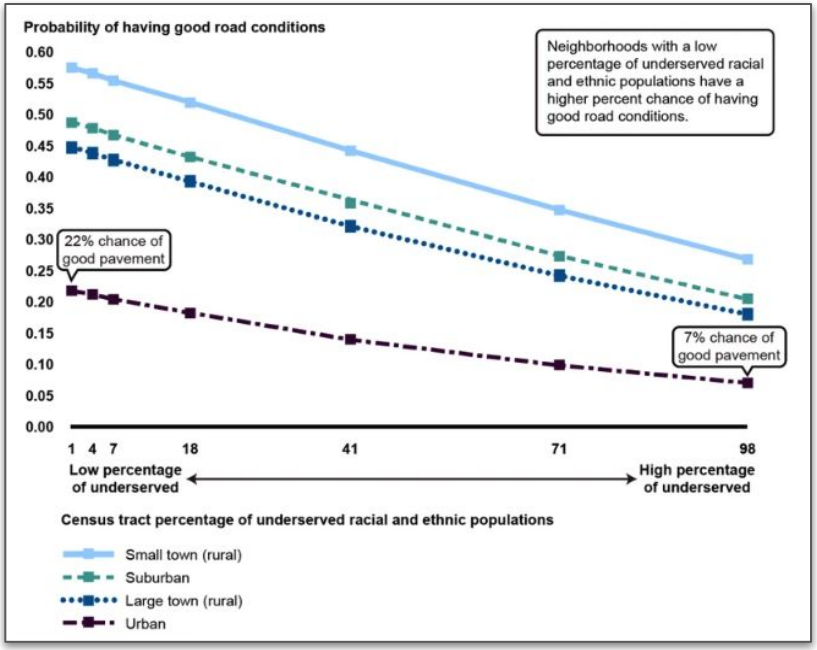
TRANSPORTATION

Poor, minority communities more likely to have bad roads, study finds

The findings highlight another challenge as the Biden administration seeks to create a more equitable transportation system

By Ian Duncan

July 29, 2022 at 8:38 p.m. EDT



https://www-washingtonpost-com.proxy.uchicago.edu/transportation/2022/07/29/road-conditions-poor-neighborhoods/
https://www.vice.com/en/article/xgyyyq/government-highways-in-poor-minority-neighborhoods-more-likely-to-be-potholed-messes

Potholes maintenance & socioeconomic status

**"The unpatched potholes are just a constant reminder of, 'Oh, of course, this is a poor neighborhood.
We're low-priority when it comes to fixing this stuff.'"**

— A resident in Oakland, California (Marketplace, 2019)

Road quality might be associated with socioeconomic status

- Report of road issue requires awareness of neighbourhood residents and certain degree of literacy
- Government response rate might differ in rich neighbourhoods and poor neighborhoods

Potholes are more concentrated in **high density residential areas** compared to low density areas (Timothy et.al, 2012)

- poor drainage on the roads, rainfall impact, advanced age of roads, poor or lack of maintenance, type and volume of traffic

CHICAGO

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Transportation

Additional Information | Related Projects | COVID Mobility Feedback | Expanded Outdoor Dining

Home / Departments / Transportation / Streets, Alleys and Sidewalks / Services / Pothole repairs

Pothole repairs

CDOT crews repair potholes on 3,800 miles of streets in neighborhoods throughout the city all year long. The months of December through April are considered the heart of the pothole season--when the city typically sees the highest number of potholes. However, crews work throughout the spring, summer and fall as well.

CDOT uses several techniques that enable the department to fill hundreds of thousands of potholes each year . They include:

- Computerized mapping and tracking of potholes reported to the City's 311 system
- Seven-day-a-week deployment of pothole repair crews during the winter months, including weekend crews and an overnight crew
- The use of high-performance cold-patch asphalt material for pothole repairs during the winter months and hot mix asphalt in spring, summer and fall.

For a comprehensive overview of CDOT's pothole operation, click on the link below:

CDOT pothole operation--winter 2010-2011--frequently asked questions

To report a pothole, click on the link below.

Get Started Online

(source: cityofchicago.org)

Potholes report procedures are complicated and require certain degree of technological literacy.

Research Question

How do Chicago's potholes being reported and being patched spatially distributed?

How are potholes associated with socioeconomic status of different areas in Chicago?

Data and Methods

Data Sources and Data Processing

- Potholes Reported - Chicago Data Portal (2021)
- Potholes Patched - Chicago Data Portal (2021)
- Census Tract Boundaries - Chicago Data Portal
- Census Tract Demographic Data - American Community Survey 5-Year Data: We used economic, housing, demographic characteristic tables from ACS, 12 useful measures were selected

The data cleaning and processing is completed with R

Research Methods

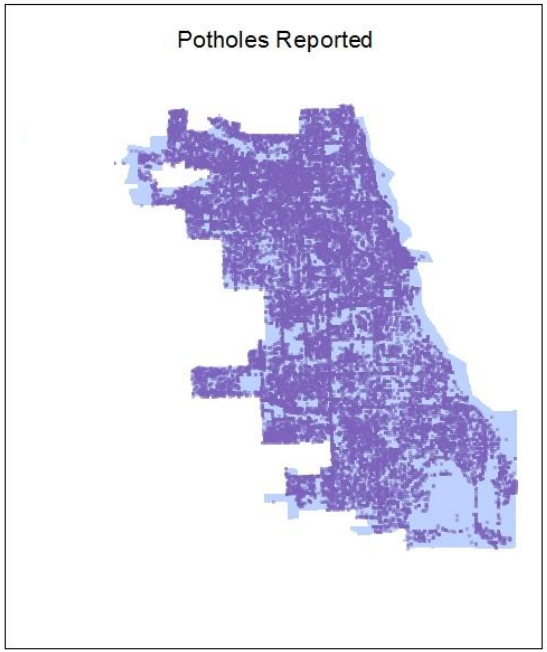
- Visualize **potholes reported/patched/ratio** by census tracts
- Visualize **demographic** information across different census tracts; Compare with pothole maps and determine how they interact with each other
- Perform **spatial autocorrelation analysis**
- Plot **scatter plots** and perform **regression analysis** to further assess the association between potholes versus different socioeconomic characteristics

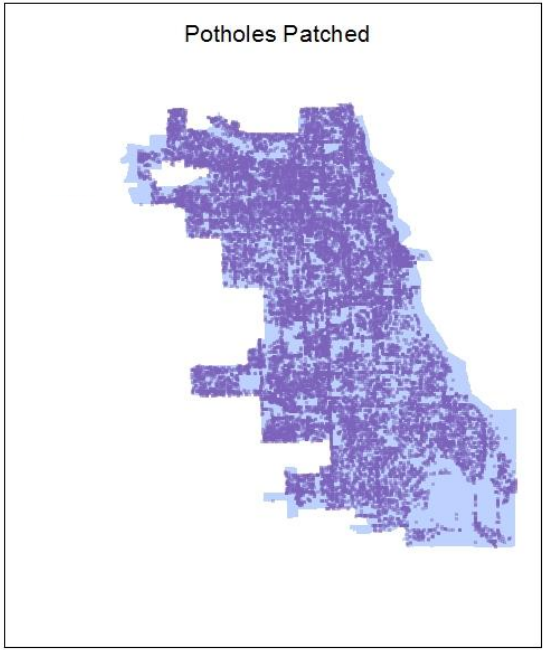
Potholes Reported & Patched in Chicago 2021

Overview of Potholes Distribution Using Raw Data

- **Need more nuanced analysis**

The Distribution of Potholes Reported and Patched in Chicago 2021



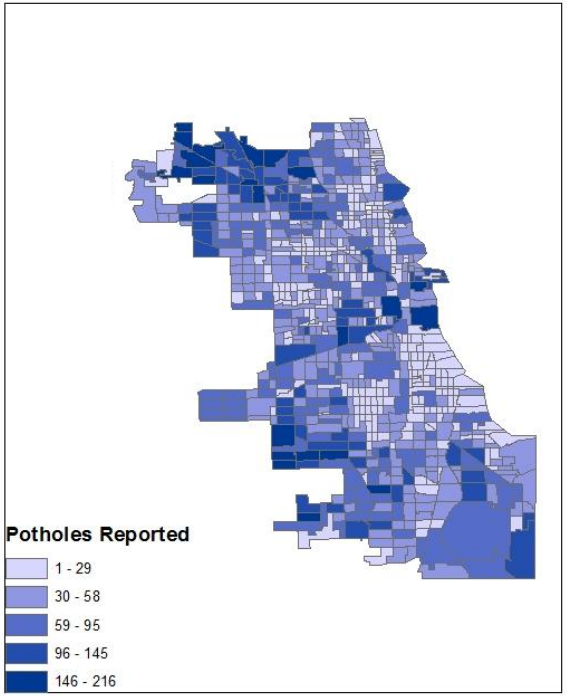


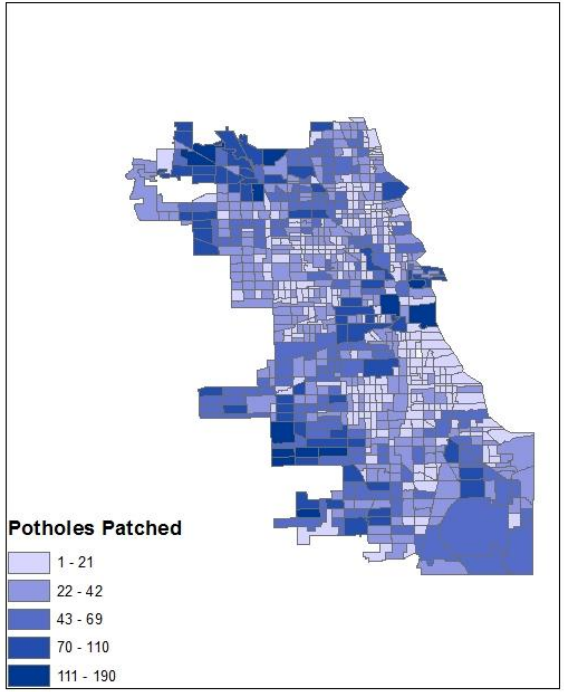
Data Source: Chicago Data Portal

Distribution of Potholes by Neighborhood

- **Disparities exist across tracts. But does population density play a role?**

The Number of Potholes Reported and Patched in Chicago 2021



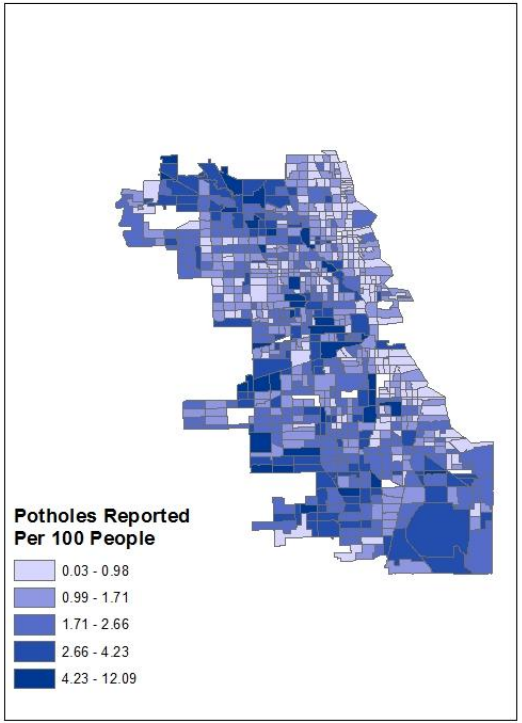


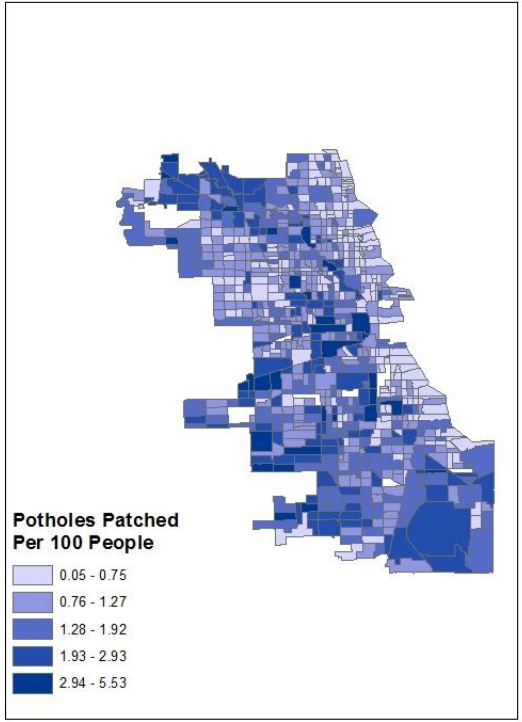
Data Source: Chicago Data Portal

Distribution of Potholes by Neighborhood (Adjusted by Population)

- Uneven potholes distribution still exist across different neighborhoods

The Number of Potholes Reported and Patched in Chicago 2021
(Standardized by Population of Age Over 16)

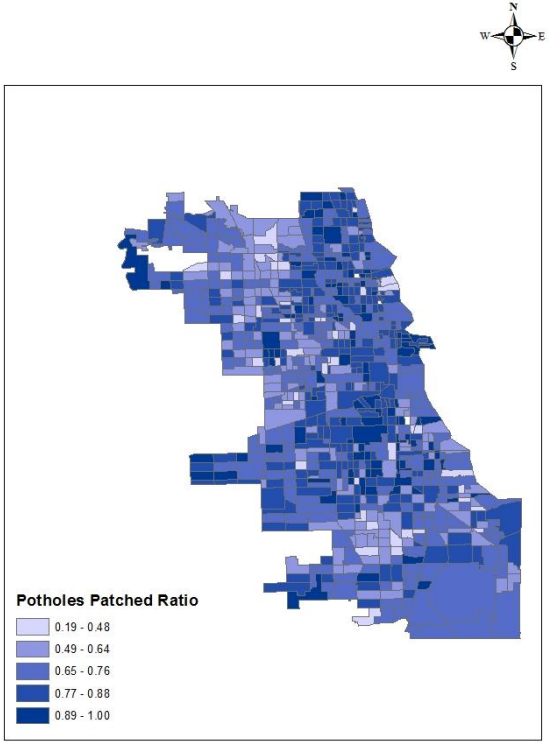




Data Source: Chicago Data Portal

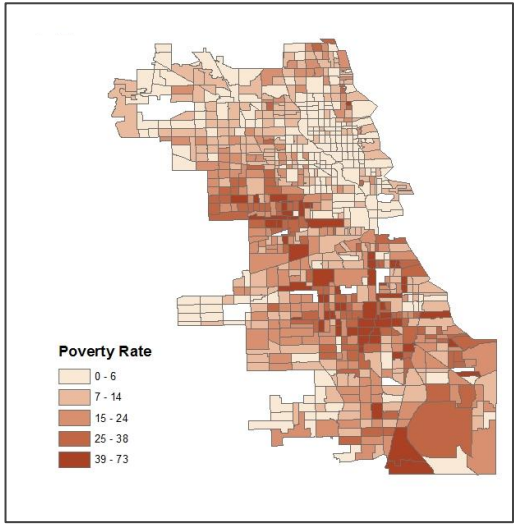
Disparities still exist across neighborhoods in whether potholes reported are actually being patched

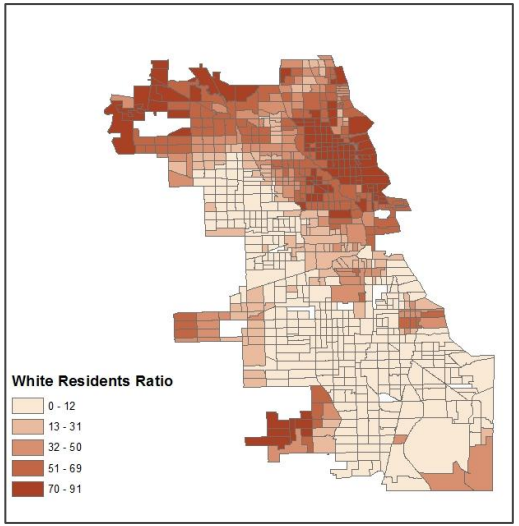
The Ratio of Patched and Reported Potholes

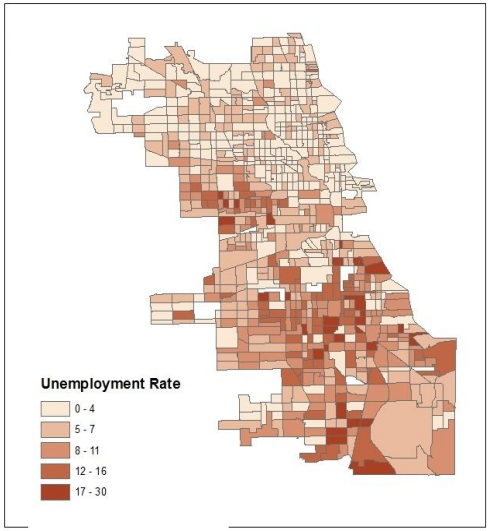


Data Source: Chicago Data Portal

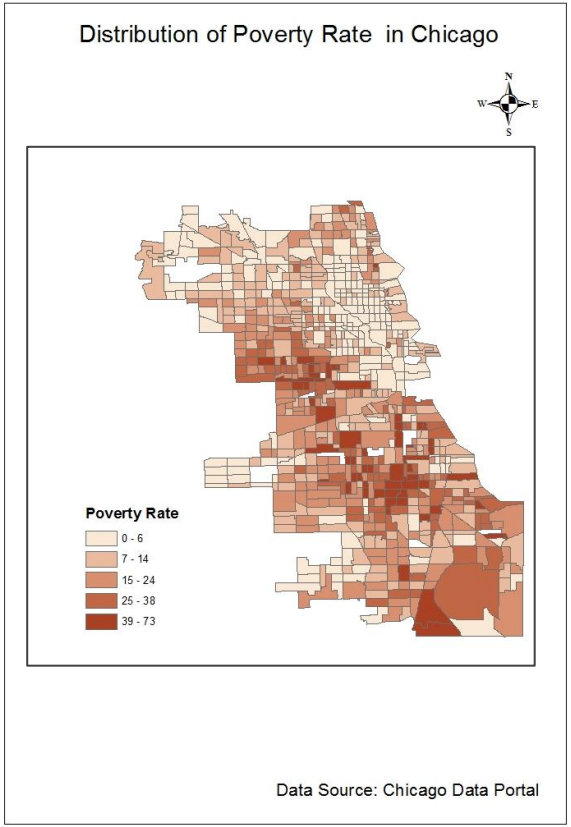
Chicago is segregated by race and poverty

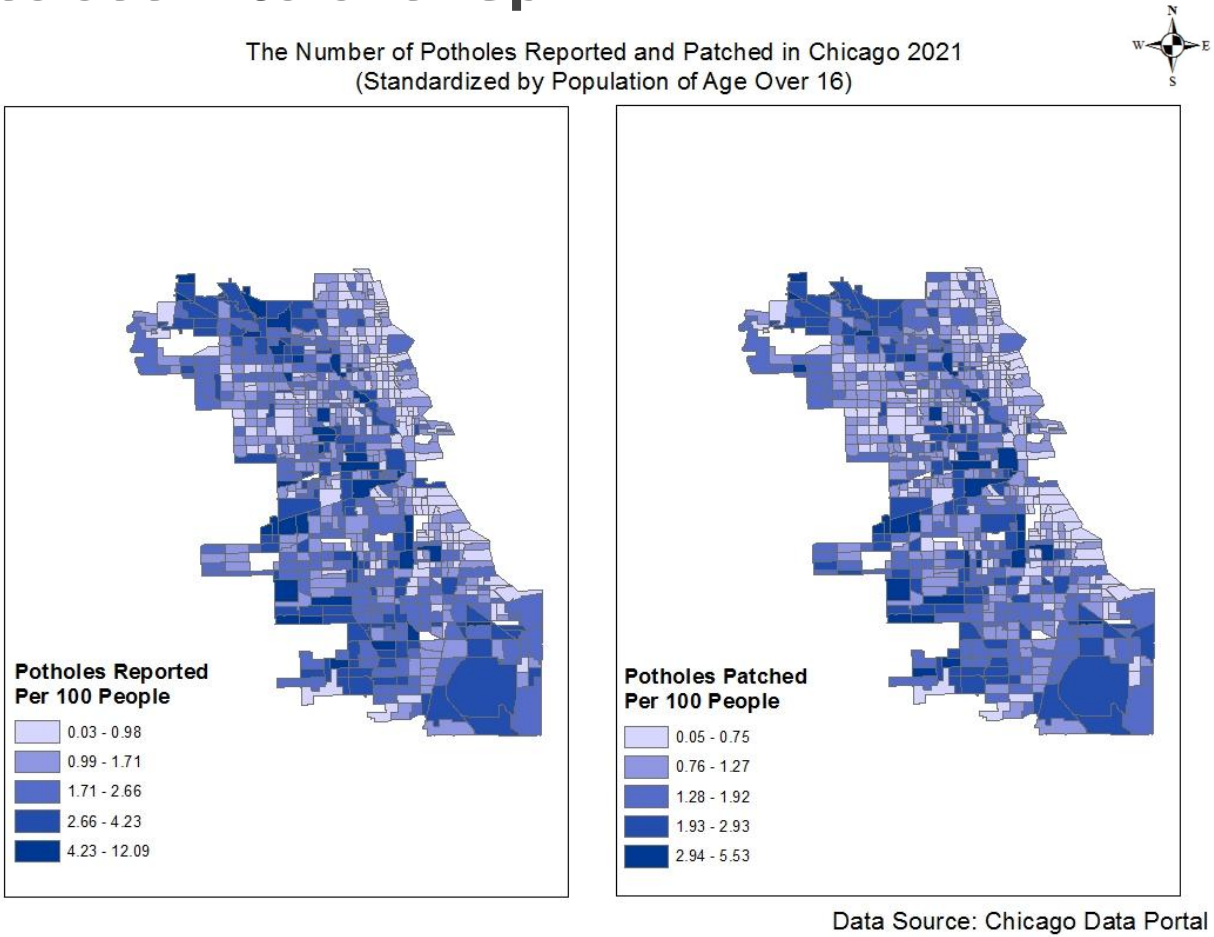




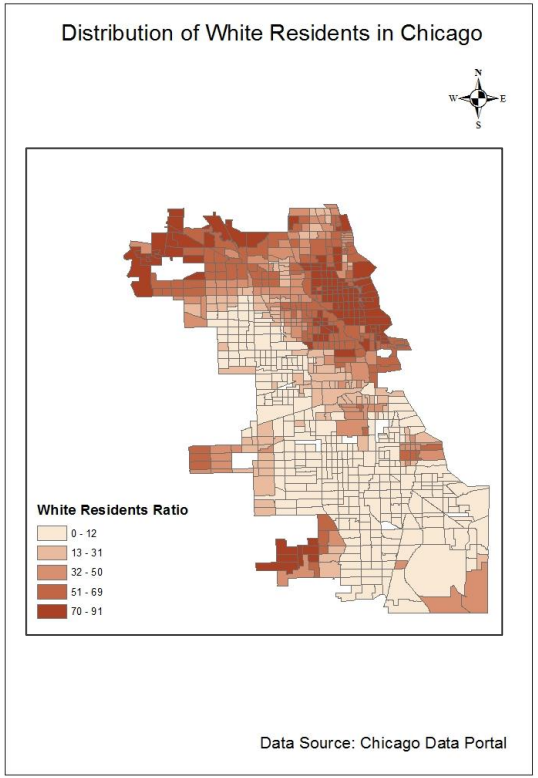


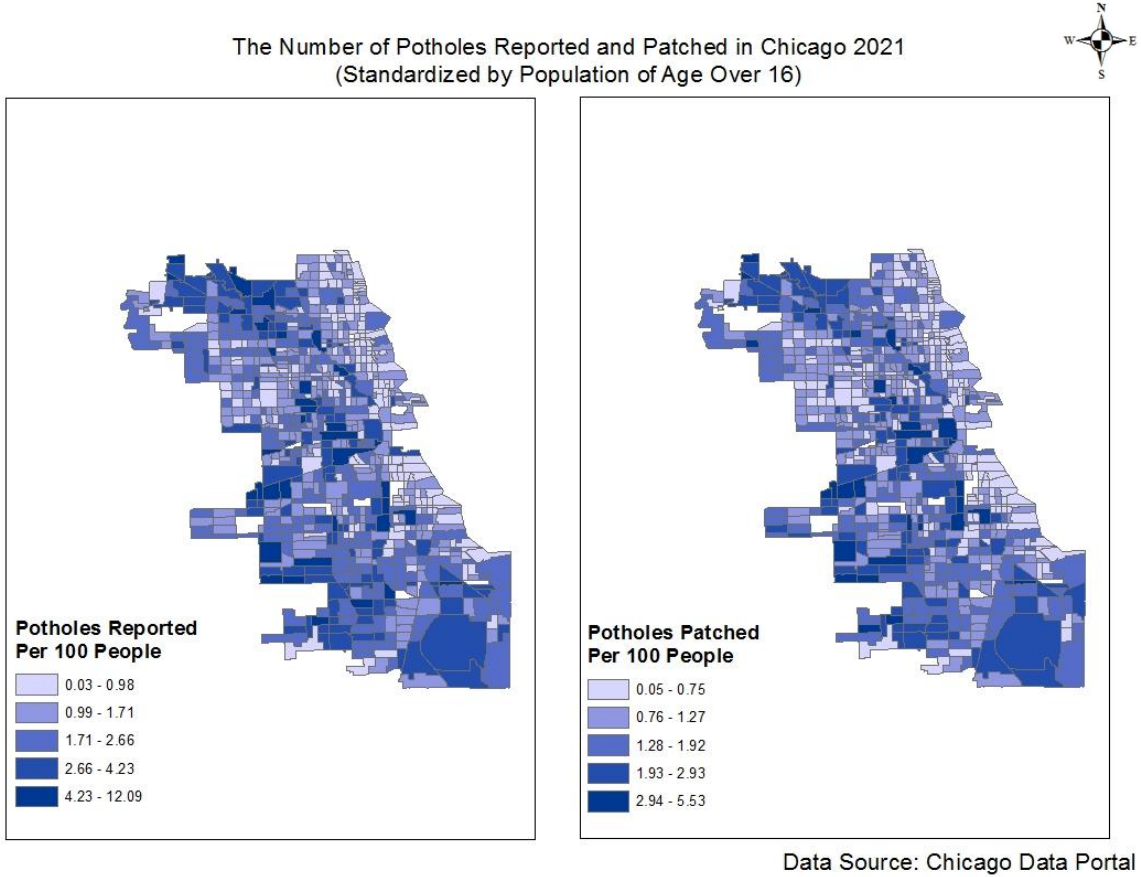
Poverty and potholes seem to overlap



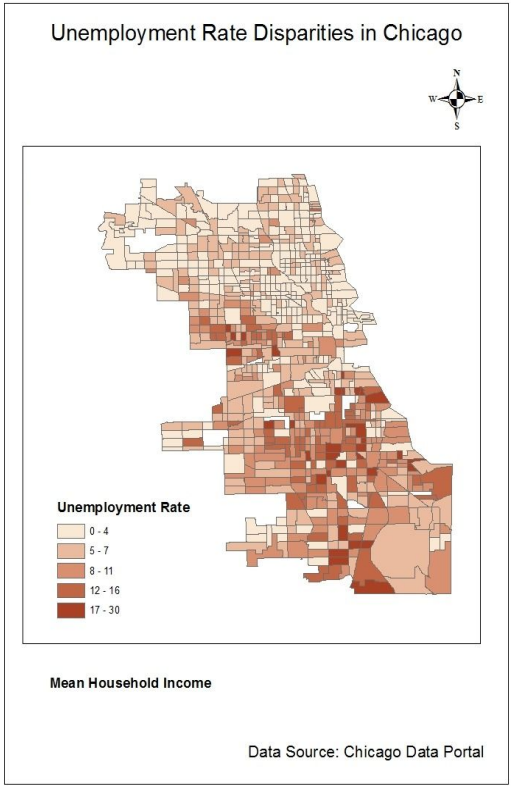


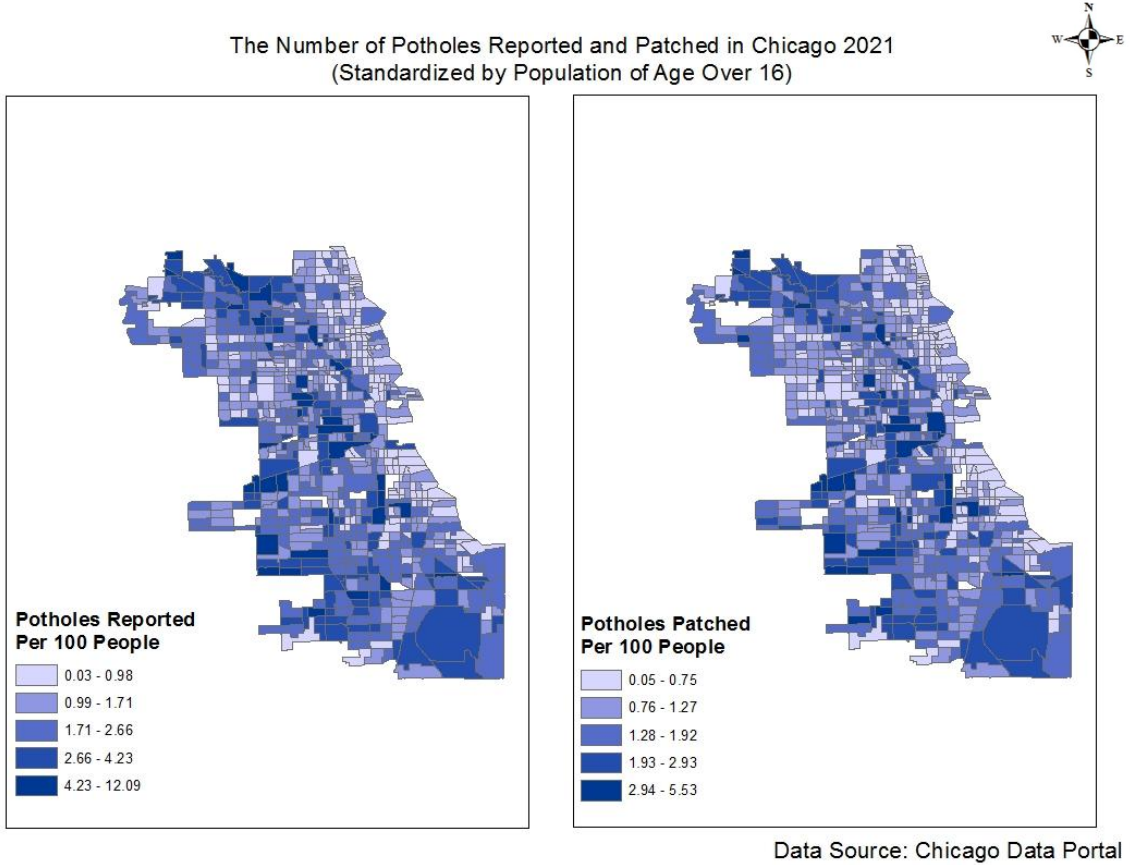
White residents condensed area seem to have less potholes





Unemployment and potholes seem to overlap





Spatial Autocorrelation & Regression Analysis

Spatial Autocorrelation

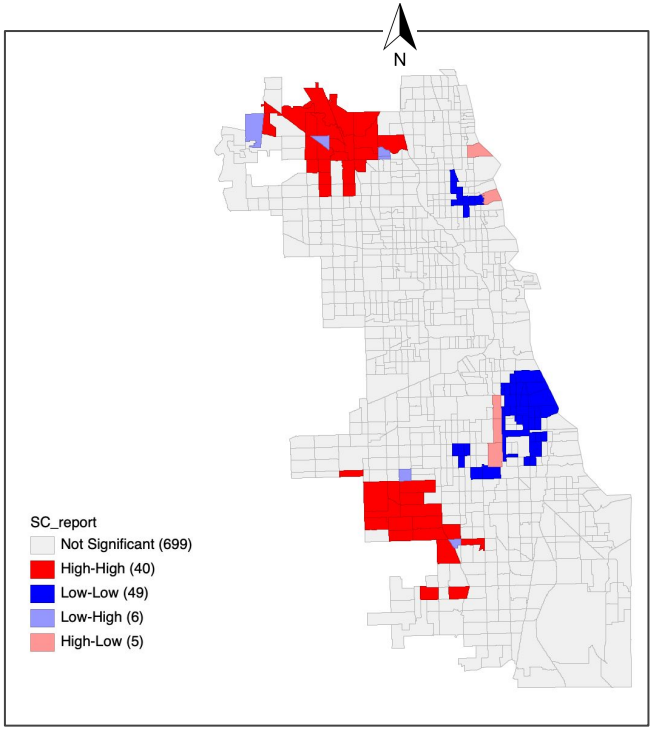
<u>Reported potholes</u>

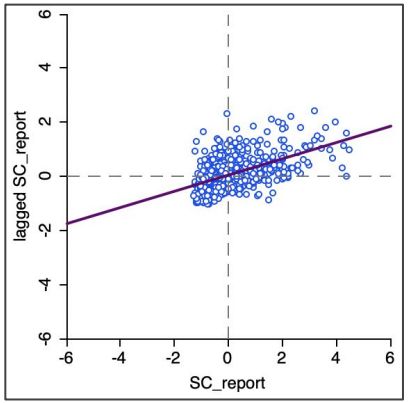
Local Moran's I Parameters:

- Second-order queen weights
- 0.001 significance level
- 99999 permutations

Red = areas with high reported potholes

Blue = areas with low reported potholes





Spatial Autocorrelation

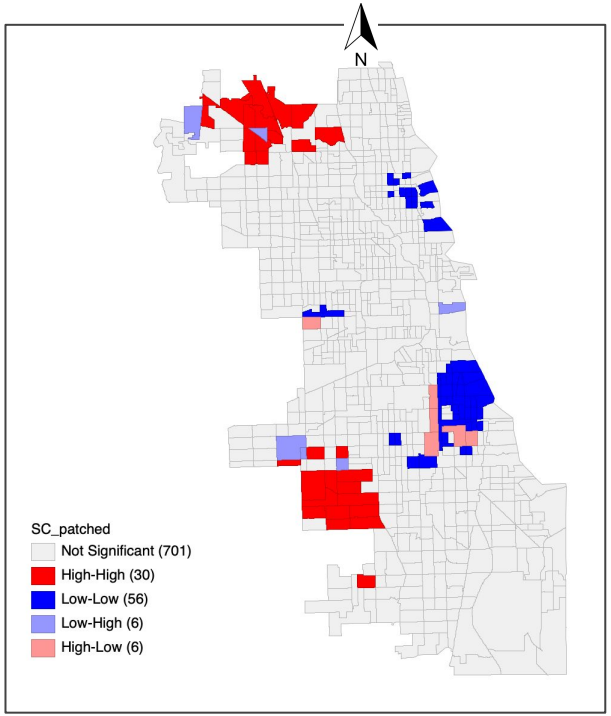
<u>Patched Potholes</u>

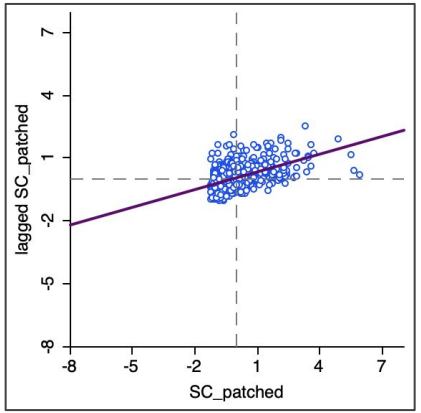
Local Moran's I Parameters:

- Second-order queen weights
- 0.001 significance level
- 99999 permutations

Red = areas with high reported potholes

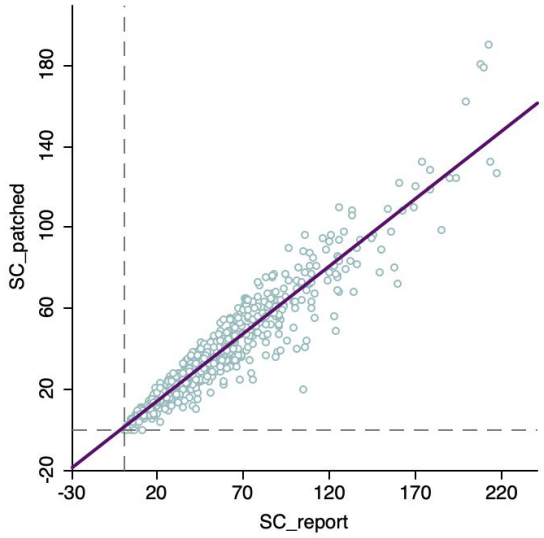
Blue = areas with low reported potholes





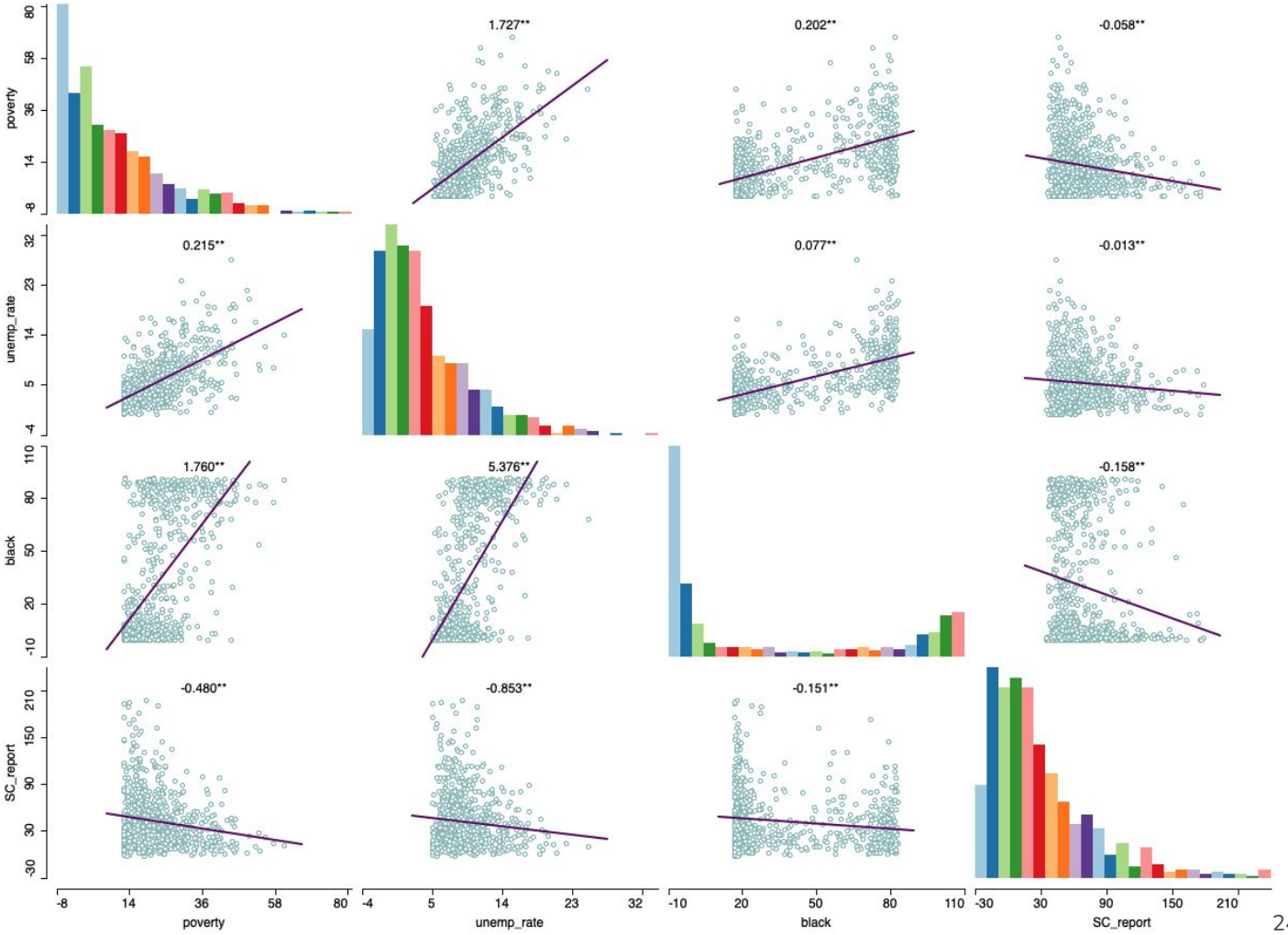
Reported vs Patched Potholes

As expected, reported and patched potholes are linearly correlated with a slope smaller than 1.



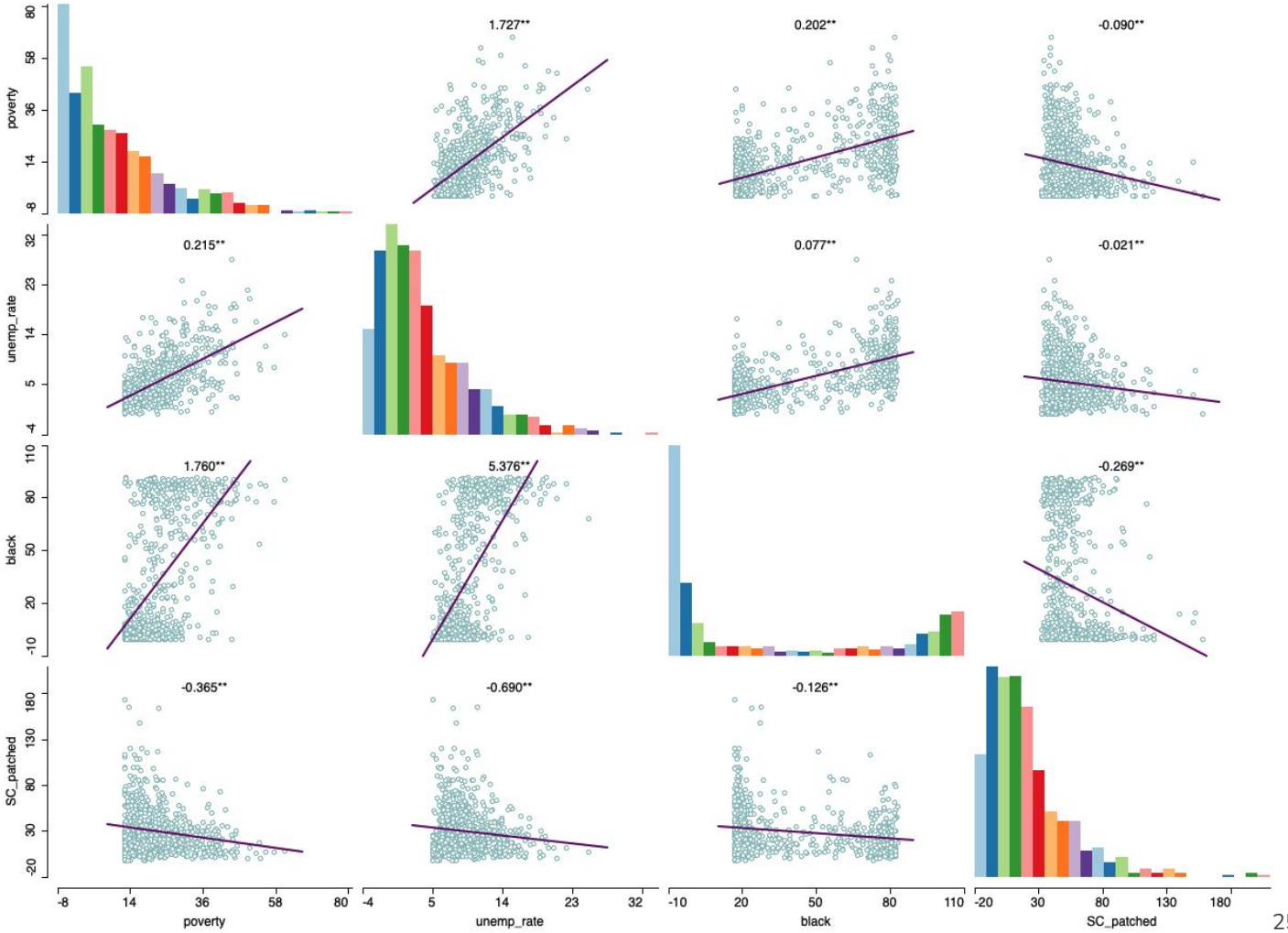
#obs	R^2	const a	std-err a	t-stat a	p-value a	slope b	std-err b	t-stat b	p-value b
801	0.912	1.579	0.446	3.542	0.000	0.667	0.007	90.904	0
0	0	0	0	0	0	0	0	0	0
801	0.912	1.579	0.446	3.542	0.000	0.667	0.007	90.904	0

The relationship between the number of **reported potholes** and socioeconomic variables



Poverty rate, unemployment rate, and percentage of African American are **negatively** & **significantly** correlated with number of reported potholes.

The relationship between the number of **patched potholes** and socioeconomic variables



Poverty rate, unemployment rate, and percentage of African American are **negatively** & **significantly** correlated with number of patched potholes.

Limitations & Next Steps

Limitation & Next Steps

We plan to analyze data from a longer timeframe.

Additional variables are required to conduct causal inference:

- It would be very helpful to gather data on the initial construction quality.
- Traffic volume could be an important factor to include.

Reference

- Mutekwa, Timothy, Mark Matsa, and Kudzanai Kanyathi. "Dodging the potholes: the spatio-distribution and socio-economic impacts of potholes in the residential areas of Gweru, Zimbabwe." (2012).
- Bradford, Ben. *One Way Oakland Is Fighting Racial Inequality? By Fixing Potholes*, May 15, 2019.
- Duncan, Lan. “Poor, Minority Communities More Likely to Have Bad Roads, Study Finds.” The Washington Post, July 29, 2022. https://www-washingtonpost-com.proxy.uchicago.edu/transportation/2022/07/29/road-conditions-poor-neighborhoods/.
- Gorden, Aaron. “Government: Highways in Poor, Minority Neighborhoods More Likely to Be Potholed Messes.” VICE, July 29, 2022. https://www.vice.com/en/article/xgyyyq/government-highways-in-poor-minority-neighborhoods-more-likely-to-be-potholed-messes.

Thank you!